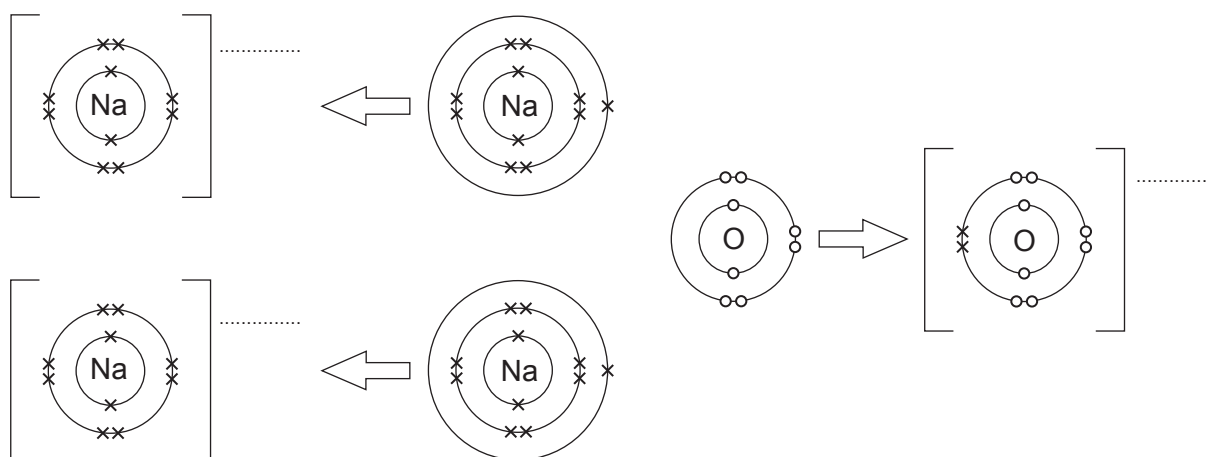


2. (a) The table gives the electronic structures of sodium and oxygen.

Element	Electronic structure
sodium	2,8,1
oxygen	2,6

(i) Sodium reacts with oxygen to give sodium oxide.

The diagram below can be used to show the electronic changes that occur as sodium oxide is formed.



Complete the diagram by

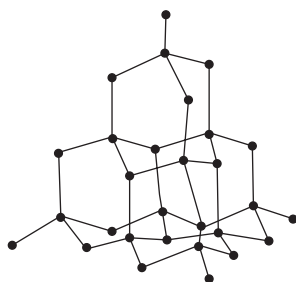
- **drawing arrows** to show the movement of electrons between the sodium and oxygen atoms,
- giving the **charges** of the sodium and oxide ions that are formed.

[2]

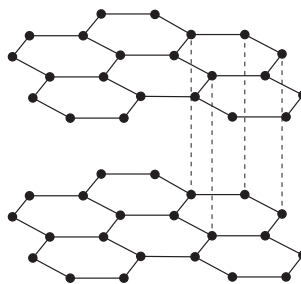


- (ii) Draw a dot and cross diagram to show the bonding in an oxygen molecule, O_2 . [2]

- (b) (i) Diamond and graphite are both made from carbon atoms. Their structures are shown below.



diamond



graphite

- I. State which structure is able to conduct electricity. Give a reason for your answer. [1]

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- II. State which structure is used as a lubricant. Give a reason for your answer. [1]

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- (ii) Give the name of another form of carbon that is used in drug delivery systems. [1]

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